



SUMULONG MEMORIAL HIGH SCHOOL
Gen. Luna St. Brgy. San Roque, Antipolo City
SY. 2022-2023

WORK, POWER AND ENERGY

LEARNING MODULE

NAME: _____

YEAR AND SECTION : _____



SET THE TARGET

LEARNING OBJECTIVES:

At the end of this module, you are expected to:

1. Describe how work related to power and energy.
2. Identify the situations in which work is done and situations.
3. Differentiate potential and kinetic.
4. Relate speed and position of object to the amount of energy possessed of a body.
5. Solve some problems related to work, energy and power

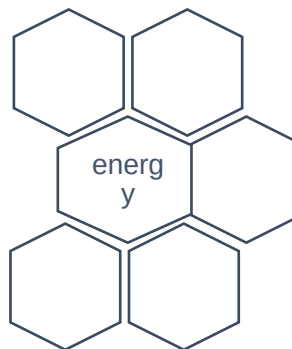
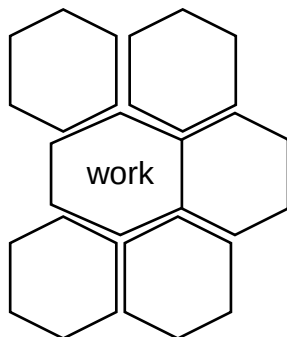
Is work being done or not?

1. Mowing the lawn
2. Weight-lifting
3. Carrying furniture from one room to another
4. Pushing against a locked door
5. Swinging a golf club
6. Hanging from a chin-up



Loop

Complete the graphic organizer below by filling up the bubbles with the words that describe work and energy. Then, answer the questions below to summarize your response.



INVESTIGATE

QUESTIONS FOR ANALYSIS

1. What can you say about your answers in the bubble map?

2. How would you compare your concept of work to that energy?

3. What did you realize after doing this activity?

Suggestion: Transfer it to activity.

I. OBJECTIVES

To understand work and its relation to power and energy.

II. MATERIALS

Balloons, golf/tennis balls, meter stick and a timer

Important: Be very careful when handling stored energy!

III. PROCEDURE

Part I

1. Release a blown-up balloon.
2. Drop a raised ball about one meter from the ground.
3. Answer the questions below under results and discussions.

Part II

1. Find the following stairway measurements: a) The height of step; and b) the number of steps to the 2nd floor. Record your measurements.
2. Climb the stairs slowly while a group mate records the time taken while climbing.
3. Repeat step 2, but this time climb faster.
4. Complete the calculation below under result and discussions.

IV. RESULT AND DISCUSSIONS

PART I

ENERGY STORED IN A BALLOON:

1. Is work done when one blows up a balloon? Explain

2. What do you call the energy stored in a blown up balloon?

3. What happens to the stored energy in the blown up balloon after you released it?

ENERGY STORED IN A BALL:

1. Is work done when one raises the ball to certain height? Explain.

2. What do you call the energy stored in a blown up balloon?

3. What happens to the stored energy after stored energy after the ball fell to the ground?

CONCLUSION

From this activity, what can you conclude about force and its relation to the object's motion?



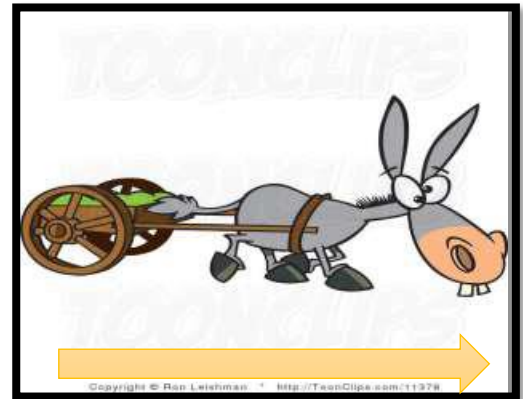
LEARN MORE!



WORK is something we do every day but have difficulty defining it. Previously, we learned the work is a quality that exists whenever the force that acts upon an object causes it to be displaced.

One Newton of Force moving 1 meter is known as a *joule* (J).

Named after British physicist **JAMES PRESCOTT JOULE**.



CALCULATING WORK

The force acts in the direction of the movement

$$\text{Work} = \text{Force} \times \text{Distance}$$

Work does NOT involve time, only force and distance.
No work is done when you stand in place holding an object.

EXAMPLE:

$$W = F \times D$$

$$\text{work} = 40\text{N} \times 0\text{m} = 0$$

Action, kid push the chair



Static friction, between feet and floor

POSITIVE WORK

In the situation you can see that the pushing force (F_{net}) is directed towards the right and that the table moves also to the right. This is a case wherein the force is completely parallel to the displacement and thus the angle is zero.

The force of the friction must be overcome first before any movement can take place.

NEGATIVE WORK

It is being pushed towards the right direction a friction force is acting on the object to the left. In this case, the value of $\theta = 180 \text{ degrees} = -1$.

The work equation now becomes

$$W_{\text{friction}} = -F_{\text{fric}} \cdot d$$

$$W_{\text{total}} = (F_{\text{applied}} - F_{\text{friction}})d$$

Work is also done vertically or against the gravity.

A pushing force
does *no* work
if the wall does
not move.



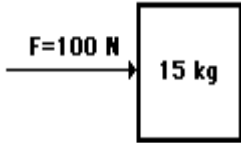
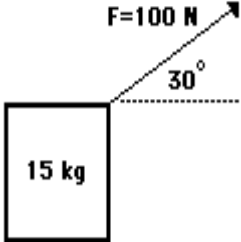
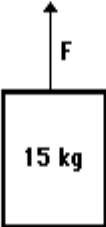
ZERO WORK

Work is scalar quantity; thus no direction is considered when adding the work done by individual forces. What if you need to raise a dumbbell from the floor to the shoulder height?



Work (force about angle to distance)

EXAMPLES

Diagram A	Diagram B	Diagram C
		
A 100 N force is applied to move a 15 kg object a horizontal distance of 5 meters at constant speed.	A 100 N force is applied at an angle of 30° to the horizontal to move a 15 kg object at a constant speed for a horizontal distance of 5 m.	An upward force is applied to lift a 15 kg object to a height of 5 meters at constant speed.

$$W = F \times d$$

$$\begin{aligned} &= 100\text{ N} \times 5\text{ m} \\ &= 500\text{ N m} \end{aligned}$$

$$W = F \times d \cos 30^\circ$$

$$\begin{aligned} &= 100\text{ N} \times 5\text{ m} \times .87 \\ &= 413\text{ N m} \end{aligned}$$

$$W = F \times d$$

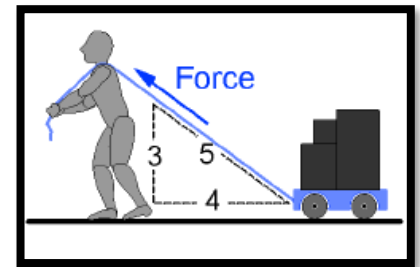
$$\begin{aligned} &= 15\text{ Kg} (10\text{ m/s}^2) \times 5\text{ m} \\ &= 750\text{ N m} \end{aligned}$$

Work (force at angle to distance)

Force (N)

Work (joules) $\rightarrow W = Fd \cos(\theta)$

Distance (m)



POWER

How quickly work is done.

Amount of work done per unit time.

$$\text{POWER} = \text{WORK} / \text{TIME}$$

POWER is simply energy exchanged per unit time, or how fast you get work done (Watts = Joules/sec)



SEE IF YOU CAN FIGURE THIS OUT

Talia and Chris have the same size yards. Talia can mow her yard in 1 hour with 30N Force, but it takes Chris two hours tomorrow his yard with 30N of force.

1. Who did more work?

2. Who used more power?

Comment: This will be part of the discussion proper.



WATT – the unit of power

A unit named after Scottish inventor **JAMES WATT**.

Invented the steam engine.

$P = W/t$ 1 watt = 1 J/s

One horsepower = 745 W

Power is the rate that we use energy.

Power = Work or Energy / Time

$P = W/t = F \times d/t = F v$

The units for power;

- J/s
- $\text{Kg m}^2 / \text{s}^2 / \text{s}$
- $\text{N m} / \text{s}$

JOULES/SECOND

WATTS

Used to measure power of light bulbs and small appliances

An electric bill is measured in kW/hrs. 1 kilowatt = 1000 W

One horsepower = 745 W

EXAMPLES

- ✚ A train pulls a load 20 meters with 3000N of force. How much work is done?

Formula

$W = F \times d$

Solution

$W = 3000 \times 20 = 60,000 \text{ Joules}$

ENERGY

POTENTIAL ENERGY is maximum at the maximum HEIGHT

After an object has been lifted to a height, work is done.

$PE = W = F \times d = mgh = 10 \text{ m/s}^2$

EXAMPLE

- ✚ How much potential energy is lost by a 5Kg object to kinetic energy due a decrease in height of 4.5 m?

Given

FORMULA = $PE = F \times D = mgh$

SOLUTION

$PE = (5\text{Kg})(10 \text{ m/s}^2)(4.5 \text{ m})$

$PE = 225 \text{ Kg m}^2/\text{s}^2$

$PE = 225 \text{ J}$

KINETIC ENERGY CALCULATION

The energy of motion

$DKE = W = F \times d = mgh = 1/2 mv^2$

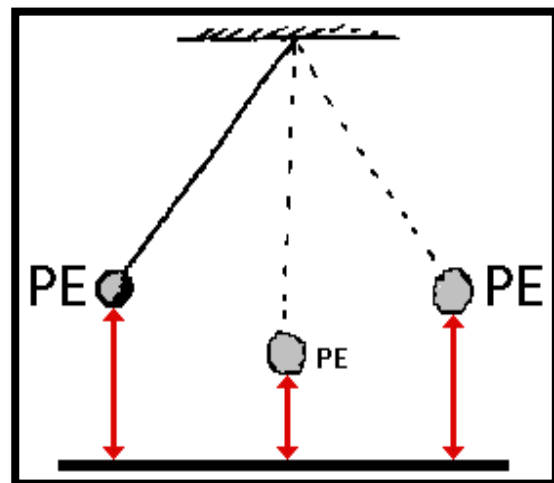
EXAMPLE

- ✚ Find the kinetic energy of an 4 Kg object moving at 5m/s.

GIVEN

FORMULA

SOLUTION



NAME: _____

YEAR AND SECTION: _____

ACTIVITY 1:

SOLVE ME!

PROCEDURE:

Answer the following word problems applying the Given, Unknown, Equation, Solution, and Answer (GUESA) pattern.

1. Alex pulls on a 15k crate with a force of 35 N at angle of 40 degrees to the horizontal How much works does he do in pulling the crate at a distance of 20m?

Given $m=15\text{kg}$, $F=35\text{N}$  $=40$ degrees, $d=20\text{m}$

Find W

Solution

2. If this kid pushed the chair across the floor 10 meters with 5N of force, how much work is done?

Formula

$$w=F \times d$$

Solution

3. When the force and displacement are perpendicular to each other, the work being done on the object is negative. When the displacement or the force is zero, no work is being done on the object. When the force is perpendicular to the displacement, the work being done is zero, no matter how large the force or the displacement.

Formula

$$w=F \times d$$

Solution

NAME: _____

YEAR AND SECTION: _____

ACTIVITY 1:

SOLVE ME!

1. A kid pushed a wagon in 30 seconds with 20 N of force. He pushed it a total of 10 meters before he gave up. **WHAT IS BEING ASKED IN THIS PROBLEM? SPECIFY.**

Given

Formula

Solution

2. A woman lifts a baby with 5N of force. She did 30J of work. How far did she lift the baby up?

Given

Formula

Solution

3. A 5 Kg Cart is pushed by a 30 N force against friction for a distance of 10m in 5 seconds. Determine the power needed to move the cart.

Given

Formula

Solution



DEFINITION OF TERMS

ENERGY is the ability to move

POTENTIAL is stored energy (Statics) and dependant on height

KINETIC is moving energy (Dynamics) and dependant on velocity

Springs store energy dependant on distance and constant

POWER is how fast the work is done

WORK is something we do every day but have difficulty defining it.

WATT – the unit of power.

WATTS -Used to measure power of light bulbs and small appliances

POWER is simply energy exchanged per unit time, or how fast you get work done
(Watts = Joules/sec)

POTENTIAL ENERGY is maximum at the maximum HEIGHT

KINETIC ENERGY is the energy of motion

Comments and suggestions:

1. Give a sample problem with atleast one derivation of formula or even units.
2. Create a question that will enhance the Higher Order Thinking Skills or **HOTS** of the students.
3. **GUESA** pattern is always applicable whenever there is a problem that is need to be solved or computed.
4. Definition of terms will be arranged alphabetically.
5. Please also include conclusion or concept micro and references electronics or printed materials see samples in other module.

